INCH-POUND
A-A-59867A
July 29, 2022
SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

HARNESS, SAFETY, FULL BODY, CROSSOVER

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) provides the requirements for a full body crossover-style safety harness for use in hazardous fall operations onboard U.S. Navy surface ships, submarines, and shore activities.

2. CLASSIFICATION. The full body crossover-style safety harness shall conform to the following sizes: X-Small (XSM), Small (SM), Standard (STD), Universal, and X-Large (XL) (see table I and figure 1).

TABLE I. Harness sizes.

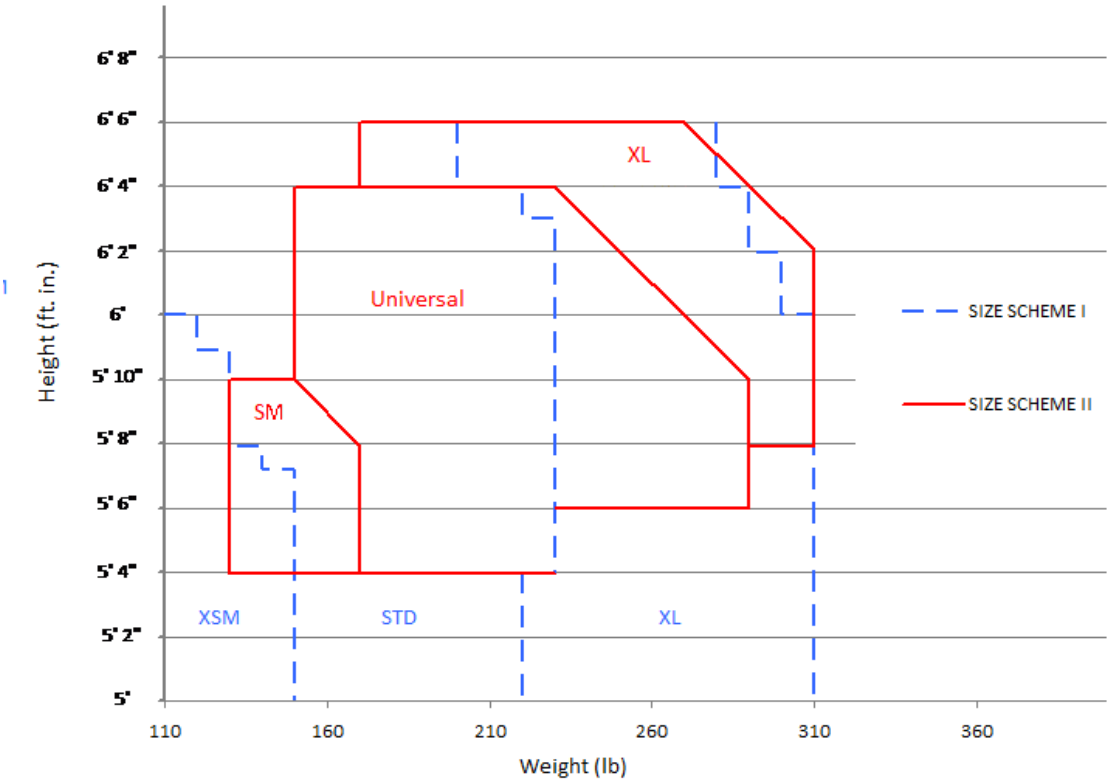
Height	**Weight (pounds)**					
	<130	**130 to <150**	**150 to <170**	**170 to <230**	**230 to <290**	**290 to 310**
Shorter than 5' 4"	XSM	XSM	STD	STD	XL	XL
Short (5' 4" to 5' 7")	XSM	XSM or SM	SM or STD	STD or Universal	Universal or XL	XL
Regular (5' 8" to 5' 11")	XSM	SM or STD	STD or Universal	STD or Universal	Universal or XL	XL
Tall (6' 0" to 6' 3")	STD	STD	STD or Universal	STD, Universal, or XL	Universal or XL	XL
Extra tall (6' 4" to 6' 6")	STD	STD	STD	STD or XL	XL	XL

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FSC 4240

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NOTES:

1. Size Scheme I: National Stock Numbers (NSN): 4240-01-545-7026 (XSM), 4240-01-544-3376 (STD), and 4240-01-544-3382 (XL).
2. Size Scheme II: NSNs: 4240-01-543-5503 (SM), 4240-01-543-5507 (Universal), and 4240-01-543-5520 (XL).

FIGURE 1. Harness sizes in graphic form.

3. SALIENT CHARACTERISTICS.

3.1 Design and construction. The full body crossover-style safety harness shall be comprised of load-bearing straps and hardware. The harness shall distribute the load from a fall over the wearer's skeletal structure. It shall be in accordance with ANSI/ASSP Z359.11, Safety Requirements for Full Body Harnesses; 29 CFR 1910, Occupational Safety and Health Standards; 29 CFR 1915, Occupational Safety and Health Standards for Shipyard Employment; and 29 CFR 1926, Safety and Health Regulations for Construction. The full body crossover-style safety harness shall have the following features:

a. The harness shall be a one-piece design that has straps across the lower chest, over the shoulders, around the thighs, and across the pelvis as shown on figure 2.

b. The harness shall be designed to allow a person to properly don and adjust an appropriate-sized harness based on the information in table I and on figure 1 in less than 3 minutes. It shall also be able to be removed readily without becoming tangled.

c. The harness shall be compatible with the U.S. Navy life preservers described in CID A-A-59767, Life Preserver, MK-1, and CID A-A-59766, Life Preserver, Inherently Buoyant with Collar.

d. The harness shall be designed for personnel weighing less than 310 pounds (140 kilograms).

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e. The harness shall have six forged D-rings. They shall be located as follows: one dorsal D-ring between the wearer's shoulder blades, one on each side at waist level, one at the chest level, and one on the top of each shoulder.

f. The harness shall have either one or two frontal pass-through buckles and color-coded straps to allow ease of use for donning. A harness with one frontal pass-through buckle shall be yellow with black or navy blue leg straps. A harness with two pass-through buckles shall be red with blue leg straps.

g. The harness shall be compatible with shock absorbing or restraint lanyards that are made in accordance with ANSI/ASSP Z359.13, Personal Energy Absorbers and Energy Absorbing Lanyards.

h. The harness shall be equipped with at least one fall arrest indicator (not shown on figure 2) that is installed and tested in accordance with ANSI/ASSP Z359.11. The fall arrest indicator shall deploy clearly and permanently after a fall and shall be located where it can be visually inspected.

i. The harness shall include at least one lanyard parking attachment element having a disengagement load of not more than 120 pounds (0.5 kilonewton). A label shall be permanently attached to the lanyard parking attachment that states, "Park Lanyard Here".

j. The harness shall have a load-bearing sub-pelvic strap.

k. The harness stretch shall not exceed 18 inches (457 millimeters) or the length stated by the manufacturer, whichever is less.

l. The harness shall have strap retainers that allow the user to control loose ends effectively.

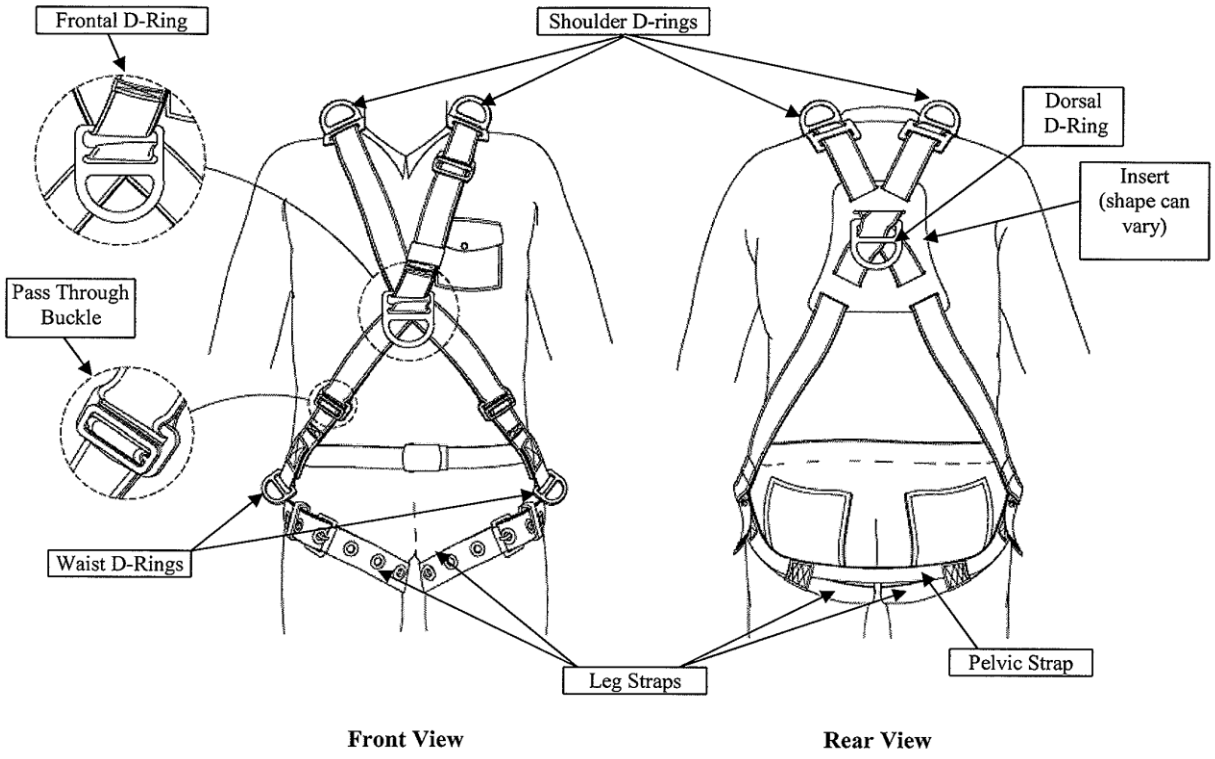


FIGURE 2. Full body, crossover safety harness (version with two front pass-through buckles shown).

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3.1.1 Hardware. The hardware or connectors shall be finished free of scale, rust, and deposits of foreign matter. The surface of the hardware shall have no burrs, pits, sharp edges, or rough surfaces. All hardware shall pass the test of ASTM B117, Standard Practice for Operating Salt Spray (Fog) Apparatus. No evidence of red rust or corrosion of the base metal shall be found after 48 hours of salt spray exposure. All hardware shall be new and in unused condition.

3.1.2 Straps. The straps shall be made from light-, heat-, aging-, and abrasion-resistant continuous virgin polyamide filament yarns or equivalent. The minimum width of the straps shall be 1⅝ inches (41 millimeters) and the strap ends shall be hot cut, sealed, covered, or stitched to prevent fraying. The straps' breaking strength shall be not less than 5,000 pounds (22.2 kilonewtons).

3.1.3 Thread and stitching. All threads shall be made of the same material as the polyamide or equivalent threads used to make the straps. The color of the threads shall contrast with the strap color for ease of inspection. All stitching shall be lock-stitched and backstitched to prevent unraveling.

3.1.4 Labeling. Labeling shall be in accordance with ANSI/ASSP Z359.11.

3.1.4.1 Legibility. The legibility and attachment of required markings shall last for the life of the harness. When pressure-sensitive labels are used, they shall be in accordance with the provisions of UL 969, Standard for Marking and Labeling Systems.

3.1.4.2 General markings. The following shall be marked in English on the equipment in accordance with ANSI/ASSP Z359.11: NSN incorporated onto an existing label, part number or model designation, month and year of manufacture, manufacturer's name or logo, identifying number "Z359.11", and a warning to follow the manufacturer's instructions included with the equipment at the time of shipment from the manufacturer.

3.1.4.3 General and specific instructions. General instructions shall be provided in accordance with ANSI/ASSP Z359.11. Specific instructions shall explain the maximum free fall distance permitted; all the markings with appropriate elaborations and explanations; and donning, fitting, adjustment, and use information including warnings against improper methods of assembly.

3.2 Performance requirements.

3.2.1 Dynamic test (dorsal attachment). The dynamic test (dorsal attachment) shall be conducted in accordance with ANSI/ASSP Z359.11.

3.2.1.1 Drop test structure. The drop structure shall allow enough vertical and horizontal clearance for the test with no obstruction. The minimum natural frequency of the drop test structure shall be 200 hertz. The maximum elastic deformation of the anchorage and test anchorage connector at the point of attachment of the test specimen shall be 0.04 inch (1 millimeter) when subjected to a vertical static load of 4,500 pounds (20 kilonewtons).

3.2.1.2 Test torso. The test drop weight shall have the form of a human torso with dimensions in accordance with ANSI/ASSP Z359.11. The torso shall be rigid, have a hardwood-like surface, and weigh 220±2 pounds (100±1 kilograms).

3.2.1.3 Test lanyard. The test lanyard shall be made from type 302 stainless steel, 7 by 19 aircraft cable construction in accordance with MIL-DTL-83420, Wire Rope, Flexible, for Aircraft Control, General Specification for. The cable diameter shall be ⅜ inch (9.5 millimeters), and lanyard eyes shall be formed by Flemish eye splices and secured with swaged fittings.

3.2.1.4 Quick release mechanism. The test torso and harness ensemble shall be released from the drop height via a remotely operated quick release mechanism that synchronously triggers the data recording device. The data for the entire history of fall arrest shall be captured.

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3.2.1.5 Testing and evaluation. The test torso and harness assembly, attached to the anchorage via the quick release mechanism, shall free fall from a height that will generate an impact load between 3,600 and 4,000 pounds (16 and 17.79 kilonewtons). After the drop, the assembly shall be suspended for 5 minutes while connected to the fall arrest system from the dorsal ring of the harness. The harness shall endure the dynamic forces of the drop and suspension period without breaking. The angle between the torso vertical centerline and vertical shall be no more than 30 degrees after the torso comes to rest. After a test fall, the fall arrest indicator shall permanently show a positive indication of deployment of the fall. A new fall arrest indicator is required at the start of each fall test. Harness stretch shall be measured and shall not exceed 18 inches (457 millimeters) or what is stated in the manufacturer's instructions, whichever is less. This test shall be conducted twice, once feet-first and once head-first. A new harness may be used for each test.

3.2.2 Dynamic test (frontal attachment). The dynamic test (frontal attachment) shall be conducted and evaluated the same way as the dynamic test (dorsal attachment) (see 3.2.1), except that the test lanyard of the fall arrest system shall be connected to the frontal D-ring of the harness and the angle between the torso vertical centerline and vertical shall be no more than 50 degrees after the torso comes to rest.

3.2.3 Strength of D-rings, O-rings, and oval rings. The strength of this hardware shall be tested in accordance with ANSI/ASSP Z359.12, Connecting Components for Personal Fall Arrest Systems. The time to reach the test load shall be greater than 1 minute to avoid dynamic effects. A static tensile test load of 5,000 pounds (22.2 kilonewtons) shall be internally applied to the hardware for a minimum period of 1 minute. The hardware shall stay intact to pass the test.

3.2.4 Strength of buckles and adjusters. Oval rings used as adjusters or other adjusters shall be subjected to the tensile testing of buckles and adjusters test in ANSI/ASSP Z359.12. The specimen shall be exposed to an internally applied test load of 3,372 pounds (15 kilonewtons). The time to reach the test load shall be greater than 1 minute to avoid dynamic effects. The test load shall be maintained for a minimum period of 1 minute and then the connector shall be evaluated. The specimen shall withstand the tensile load without breaking to pass the test.

3.2.5 Proof loading of D-rings and oval rings. Connectors shall be subjected to 100-percent proof load testing in accordance with ANSI/ASSP Z359.12. The time required to reach 3,600 pounds (16 kilonewtons) proof load shall be greater than 1 second. The proof load shall be maintained for a period of not less than 2 seconds. Inspection after testing shall not reveal any cracking, breaking, or permanent deformation visible to the unaided eye.

3.2.6 Tensile strength of straps. Straps shall have a breaking strength of not less than 5,000 pounds (22.2 kilonewtons) when tested in accordance with FED-STD-191, Textile Test Methods, method 4108, Strength and Elongation, Breaking; Textile Webbing, Tape and Braided Items. Straps shall maintain a breaking strength of not less than 3,600 pounds (16 kilonewtons) after abrasion conditioning in accordance with FED-STD-191, method 5309, Abrasion Resistance of Textile Webbing.

3.2.7 Static strength of the full body harness. The standard 220±2-pound (100±1-kilogram), torso-shaped, hardwood-like surface rigid test weight shall be properly fitted in the full body harness in accordance with ANSI/ASSP Z359.11. The assembly shall then be attached to the lower and upper jaws of the testing machine by the torso crotch ring and harness dorsal D-ring, respectively. The location of the buckles and adjusters shall be marked before testing in order to measure slippage later. The load shall be increased up to 3,600 pounds (16 kilonewtons) at a rate of 2 inches per minute (51 millimeters per minute) and maintained for 1 minute. The harness shall hold the test torso during the test and no element should break. Adjusters shall not slip more than 1 inch (25 millimeters). Buckle and eyelet type of construction shall not tear a distance greater than that to the adjacent eyelet. Except for the straps of buckle and eyelet adjusters, straps shall not show any sign of tearing.

3.2.8 Static strength of full body harness (front-mounted attachment). The same test for static strength of the full body harness shall be used. The same procedures to make a pass or fail decision shall be followed.

4. REGULATORY REQUIREMENTS. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. Manufacturers shall show that they have made similar harnesses that meet ANSI/ASSP Z359.11, 29 CFR 1910, 29 CFR 1915, and 29 CFR 1926.

5.3 First article. One full body crossover harness shall be subjected to first article inspection in accordance with 3.1 and 3.2. Results shall be provided as specified in the contract or order (see 7.2).

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 ANSI standards. ANSI standards are available from https://webstore.ansi.org/ or the American National Standards Institute, 25 W. 43rd Street, 4th Floor, New York, NY 10036.

7.1.2 ASTM standards. ASTM standards are available from www.astm.org or ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959.

7.1.3 Code of Federal Regulations (CFR). The CFR is available from www.ecfr.gov or the Superintendent of Documents, U.S. Government Publishing Office, Washington, DC 20401.

7.1.4 Defense specifications, Federal standards, and CIDs. Defense specifications, Federal standards, and CIDs are available online at https://quicksearch.dla.mil.

7.1.5 UL standards. UL standards are available from www.ul.com/ or COMM 2000, 151 Eastern Avenue, Bensenville, IL 60106.

7.2 Ordering data. The contract or order should specify the following:

a. CID document number and revision.
b. NSNs.
c. Product conformance provisions, including first article test results (see 5.3).
d. Quantity required.
e. Packaging requirements.

7.3 National stock numbers (NSN). The following is a list of NSNs assigned corresponding to this CID. The list may not be indicative of all possible NSNs associated with this CID.

NSN	ITEM
4240-01-545-7026	Size X-Small Safety Harness
4240-01-543-5503	Size Small Safety Harness
4240-01-544-3376	Size Standard Safety Harness
4240-01-543-5507	Size Universal Safety Harness
4240-01-544-3382 or 4240-01-543-5520	Size X-Large Safety Harness

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7.4 Key words.

Fall protection
Hardware
Straps

CONCLUDING MATERIAL

MILITARY INTERESTS

Custodians:
Army – EA
Navy – SH
Air Force – 99

Review Activities:
Army – TE
Navy – AS
Air Force – 03, 184

Preparing Activity:
Navy – SH
(Project 4240-2022-002)

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